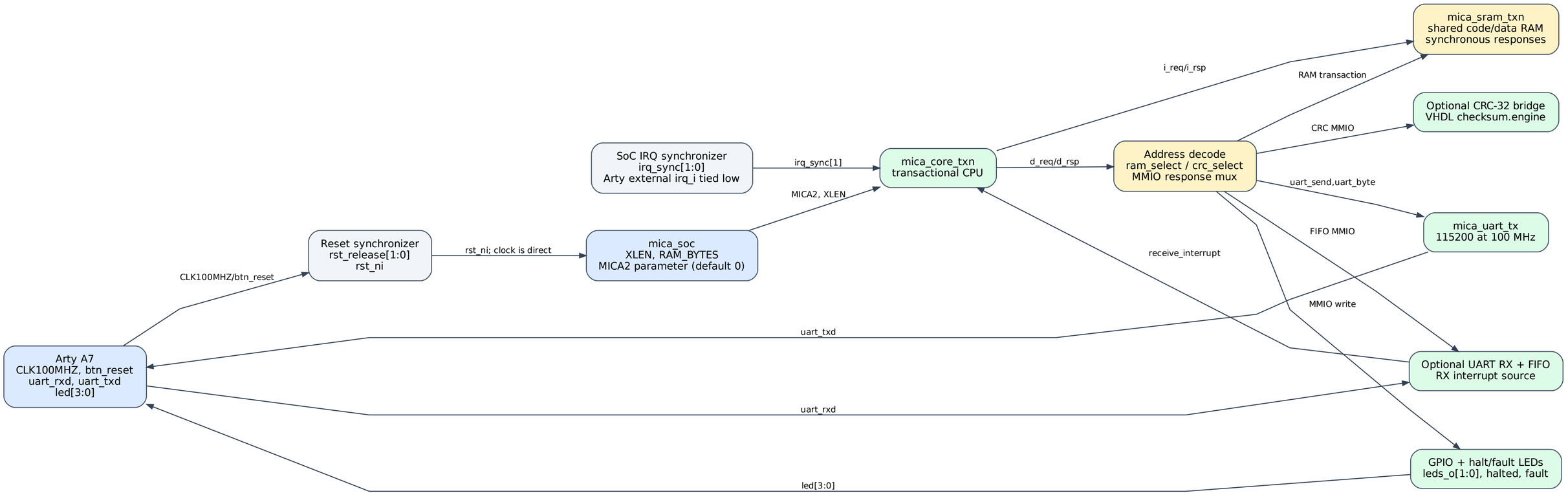
MICA 2.0 | SoC wiring and board-level boundary
Arty A7
CLK100MHZ, btn_reset
uart_rxd, uart_txd
led[3:0]
CLK100MHZ/btn_reset
Reset synchronizer
rst_release[1:0]
rst_ni
rst_ni; clock is direct
mica_soc
XLEN, RAM_BYTES
MICA2 parameter (default 0)
MICA2, XLEN
SoC IRQ synchronizer
irq_sync[1:0]
Arty external irq_i tied low
irq_sync[1]
mica_core_txn
transactional CPU
i_req/i_rsp
d_req/d_rsp
Address decode
ram_select / crc_select
MMIO response mux
RAM transaction
mica_sram_txn
shared code/data RAM
synchronous responses
CRC MMIO
Optional CRC-32 bridge
VHDL checksum.engine
uart_send,uart_byte
mica_uart_tx
115200 at 100 MHz
FIFO MMIO
receive_interrupt
MMIO write
Optional UART RX + FIFO
RX interrupt source
uart_txd
uart_rxd
GPIO + halt/fault LEDs
leds_o[1:0], halted, fault
led[3:0]